

An Analysis of Gold and Bitcoin as Safe Havens and Inflation Hedges

Abstract

While the defensive roles of Gold and Bitcoin were heavily debated during the COVID-19 era, their comparative efficacy in the post-pandemic "New Normal"—marked by the "Trump 2.0" transition—requires fresh empirical scrutiny. This study investigates the performance of Gold and Bitcoin as safe havens and inflation hedges between 2023 and 2026, specifically examining their response to forward-looking inflation expectations (T5YIFR) and market volatility (VIX). Our empirical results reveal a nuanced divergence from traditional expectations. During high-volatility regimes ($VIX > 18$, $n=108$), Gold demonstrated structural resilience with a near-zero correlation (0.071) to the S&P 500, functioning as a robust diversifier that decouples from equity drawdowns. In contrast, Bitcoin exhibited a moderate positive correlation (0.369), suggesting that its market behavior increasingly aligns with pro-cyclical risk assets rather than traditional safe havens in the current macroeconomic climate. Furthermore, high-frequency analysis of inflation hedging capabilities indicates a significant attenuation of immediate responsiveness; both Bitcoin (0.003) and Gold (-0.029) showed marginal sensitivity to daily changes in the 5-Year, 5-Year Forward Inflation Expectation Rate. These findings suggest that while these assets may maintain long-term equilibrium with inflation, their short-term hedging efficacy is limited under current monetary conditions. The study concludes that in the current macroeconomic environment, Gold remains the primary instrument for portfolio stabilization through non-correlation, while Bitcoin's role as "digital gold" is constrained by its increasing integration with broader risk-on market dynamics.

Keywords: Bitcoin; Gold; Safe-Haven Assets; Inflation Expectations; DCC-GARCH

1. Introduction

The global financial landscape in the mid-2020s has been characterized by unprecedented volatility, driven by persistent inflationary pressures, shifting geopolitical alliances, and the structural economic transformations of the "Trump 2.0" era (Julaiti et al., 2026). In this turbulent environment, the search for reliable "Safe Havens" (assets that retain or gain value during market crashes) and "Inflation Hedges" (assets that protect purchasing power) has become a primary concern for both institutional and retail investors. Traditionally, Gold has reigned as the ultimate store of value for centuries. However, the rapid institutionalization of Bitcoin, often dubbed "Digital Gold," has introduced a provocative alternative into the modern portfolio.

Despite the growing popularity of Bitcoin, its defensive efficacy remains a subject of intense academic and practical debate. While early proponents argued that Bitcoin's fixed supply makes it an ideal hedge against fiat debasement, recent market episodes have shown a troubling "stress-selectivity"(Teruel-Gutiérrez et al., 2026). Unlike Gold, which remains relatively insulated from fear-driven volatility spikes, Bitcoin has frequently exhibited high correlations with risky assets during systemic liquidity crises. This "Safe-Haven paradox" suggests that the roles of these two assets are not static but evolve across different frequencies and crisis archetypes (Li & Yin, 2026).

This report aims to conduct a comprehensive empirical analysis of Gold and Bitcoin's performance as safe havens and inflation hedges between 2023 and 2026. The specific objectives are: To evaluate the dynamic correlation between Gold/Bitcoin and global equity markets during distinct stress regimes (valuation-driven vs. uncertainty-driven). To analyze the effectiveness of both assets in hedging against realized and expected inflation in the current high-interest-rate environment. To investigate how institutional adoption and political shifts have altered Bitcoin's operational mechanism compared to Gold's intrinsic value preservation.

The remainder of this report is organized as follows: Chapter 2 provides a critical review of the existing literature, spanning from early COVID-19 studies to the latest 2026 findings. Chapter 3 outlines the methodology, including the data collection process and the econometric models (DCC-GARCH) employed for analysis. Chapter 4 presents empirical results, highlighting the significant divergence in performance between Gold and Bitcoin. These findings are further analyzed and interpreted in the discussion in Chapter 5. Finally, Chapter 6 concludes with practical implications for portfolio management and policy-making.

2. Literature Review

The conceptualization of safe havens was traditionally dominated by Gold, owing to its scarcity and lack of counterparty risk. Early empirical studies during the COVID-19 pandemic, initially positioned Bitcoin as a viable digital counterpart to Gold (Dwita Mariana et al., 2021; Wen et al., 2022). These authors observed that during the initial liquidity shocks of 2020, both assets provided varying degrees of diversification. However, even within this period, a "Safe-Haven Paradox" began to emerge; Li notes that Gold's defensive utility occasionally collapsed during extreme "dash for cash" episodes, suggesting that even traditional safe havens are susceptible to liquidity-driven systemic shocks (Li & Yin, 2026).

A significant fracture in the literature exists regarding whether Bitcoin qualifies as a true inflation hedge or merely a speculative vehicle. Choi argues that while Bitcoin responds to inflation expectations, it lacks the stable "store of value" characteristics inherent in Gold, acting more as a high-beta risk asset than a hedge (Choi & Shin, 2022). This skepticism is echoed by Acikgoz, who employs multifractal cross-correlation analysis to reveal that neither asset consistently functions as a safe haven across all asset classes, particularly when markets exhibit nonlinear dynamics and inefficiency (Acikgoz, 2025). These findings suggest that the hedging efficacy of Bitcoin may be an episodic phenomenon rather than an intrinsic property.

As the global economy transitioned from the pandemic to geopolitical conflicts, researchers began to emphasize "contextual heterogeneity." Manzli expanded the research scope to include the Russia-Ukraine conflict and the Silicon Valley Bank (SVB) collapse, discovering that Gold consistently outperforms Bitcoin during banking-sector crises (Snene Manzli & Jeribi, 2024). Interestingly, Manzli also highlights the emergence of gold-backed cryptocurrencies as a superior hybrid instrument, offering the regulatory transparency of blockchain with the stability of bullion. This evolution indicates that the "Digital Gold" narrative is shifting toward more stable, asset-backed digital architectures.

The most recent literature focuses on the structural shifts in the current macroeconomic environment and the nuanced nature of market stress. Julaiti contends that the institutionalization of Bitcoin under evolving U.S. regulatory frameworks has decoupled its behavior from Gold; while Gold relies on intrinsic value, Bitcoin's role is increasingly defined by its market isolation and institutional flow (Julaiti et al., 2026). Crucially, Teruel-Gutiérrez provides a definitive refinement by distinguishing between "valuation-driven" and "uncertainty-driven" stress (Teruel-Gutiérrez et al., 2026). Teruel-Gutiérrez finds that Bitcoin is "stress-selective"—it may offer diversification during standard market corrections but tends to fail during fear-driven volatility spikes where its correlation with equities surges. This distinction resolves the mixed evidence found in earlier studies and provides a robust framework for evaluating these assets in the current 2026 economic climate.

While the existing literature has extensively covered the 2020-2024 period, there remains a gap in evaluating the long-term convergence or divergence of these assets as we enter 2026. Most studies focus on either short-term volatility or long-term hedging, but few integrate the "stress-selective" framework with the latest institutional shifts of the current macroeconomic environment.

This report seeks to fill this gap by providing a frequency-specific analysis of Gold and Bitcoin in the current multi-polar financial order.

3. Methodology

3.1 Data description and sources

To provide a robust empirical analysis, this study utilizes a multi-asset dataset spanning the critical period of 2023–2026. Given the forward-looking nature of financial markets, we emphasize market-based expectations over lagging indicators. In this study, both traditional and digital safe-haven assets are considered, with the LBMA AM USD Gold Price serving as the conventional benchmark and the Coinbase Bitcoin (CBBTCUSD) price representing its digital counterpart. The S&P 500 Index is employed as a proxy for overall US equity market performance, while market stress and investor sentiment are captured through the CBOE Volatility Index (VIX). To assess inflation expectations, the analysis utilizes the 5-Year, 5-Year Forward Inflation Expectation Rate (T5YIFR) derived from the Inflation-Linked Swap market. Unlike the lagging Consumer Price Index (CPI), this measure reflects the market’s forward-looking anticipation of long-term monetary debasement, providing a more timely indicator of inflation expectations in financial decision-making.

Table 1. Assets and Indicators for Safe-Haven and Inflation-Hedging Analysis

Asset/Indicator	Frequency	Primary Source	Description
Gold	Daily	London Bullion Market Association	The global benchmark price for gold, denominated in USD per troy ounce.
Bitcoin	Daily	Coinbase (via FRED/Database)	The USD price of Bitcoin, representing the primary global cryptocurrency asset.
Equities	Daily	S&P Dow Jones Indices	The benchmark for the US large-cap equity market.
Market Stress	Daily	CBOE Volatility Index	Known as the "Fear Gauge," representing US equity market volatility.
Inflation	Monthly (forward-filled to daily)	FRED / LSEG	5-Year, 5-Year Forward Inflation Expectation Rate

3.2 Data Pre-processing

Before conducting the empirical analysis, several pre-processing steps are implemented to ensure data consistency. To address the non-stationarity of price levels and allow for comparison across different scales, all daily prices are converted into logarithmic returns. Logarithmic returns are preferred in this study over simple returns to ensure time-additivity and to mitigate the impact of heteroskedasticity, which is particularly prevalent in highly volatile assets like Bitcoin.

$$r_{i,t} = \ln(P_{i,t}/P_{i,t-1}) \times 100 \quad (1)$$

where $r_{i,t}$ represents the return of asset i at time t ; $P_{i,t}$ defines the closing price of asset i on the observation day t ; $P_{i,t}/P_{i,t-1}$ defines the price change ratio. If this value is greater than 1, it indicates the asset has increased in price; if less than 1, it indicates the asset has decreased in price.

Following this, inflation expectation shifts are calculated: since inflation swaps are expressed in percentages, we compute the daily first-difference (ΔInf_t) to capture the "shocks" in inflation sentiment, which serves as a dynamic measure of changes in market expectations over time.

$$\Delta Inf_t = Inf_Swap_t - Inf_Swap_{t-1} \quad (2)$$

This allows us to analyze how Gold and Bitcoin react to immediate changes in inflation expectations.

A significant challenge in this empirical study is the inherent frequency mismatch between the macroeconomic indicator and market-based financial data. Specifically, the T5YIFR obtained from LSEG is a monthly series, reflecting long-term structural expectations. In contrast, the prices of Gold, Bitcoin, and the S&P 500, along with the VIX Index, are recorded at a daily frequency. To ensure the econometric robustness of the DCC-GARCH model—which requires a high-frequency observation density to capture volatility clustering—we implemented a Forward-Filling Interpolation (also known as a constant-attribute resampling technique) for the T5YIFR series. This methodology proceeds under the rational assumption that forward-looking inflation expectations represent the "market consensus" and remain functionally static in the eyes of investors until a new monthly observation or macroeconomic catalyst is released. Furthermore, to maintain statistical stationarity for the multivariate

framework, we utilized the first difference (Δ) of the resampled daily inflation expectations. This alignment strategy allows the model to preserve the high-frequency dynamics of asset price returns while incorporating inflation expectations as a consistent exogenous signal. This approach minimizes data loss that would otherwise occur through down-sampling and ensures that the conditional correlations are estimated over a sufficiently large sample size to guarantee asymptotic efficiency.

3.3 Econometric Framework

3.3.1 Safe-Haven Testing (DCC-GARCH Approximation)

We evaluate the time-varying correlation between the safe-haven candidates and the S&P 500. A "safe-haven" is defined as an asset that exhibits a non-positive correlation with equities during periods of extreme market stress. We specifically test for "stress-selectivity" by comparing correlations in normal regimes versus high-volatility regimes ($VIX > 18$).

3.3.2 Inflation-Hedge Regression

To determine the hedging efficacy against inflation expectations, we employ an OLS regression model with HAC-robust standard errors:

$$R_{Asset,t} = \alpha + \beta_1 R_{SP500,t} + \beta_2 \Delta VIX_t + \beta_3 \pi_t + \epsilon_t \quad (3)$$

where π_t represents the inflation proxy. This model controls market movements and volatility shocks to isolate the intrinsic defensive utility of gold versus bitcoin. β_3 represents the inflation hedging coefficient. A statistically significant positive β_3 indicates that the asset successfully protects the investor's purchasing power against shifts in long-term inflation sentiment.

3.3.3 Visual and regime analysis

The analysis is supplemented by visual inspections of normalized price performance and a frequency-based comparison of correlations. This multi-layered approach ensures that the findings are robust against both valuation-driven drawdowns and uncertainty-driven volatility spikes.

4. Results

4.1 Comparative performance and volatility

Figure 1 presents the normalized price performance of Bitcoin, Gold, and the S&P 500 over the period from 2023 to 2026. Bitcoin's normalized price rises to approximately 400, significantly outperforming Gold (~240) and the S&P 500 (~155). In contrast, Gold demonstrates a steady upward trend with relatively low fluctuations, highlighting its role as a stable store of value. While

Bitcoin's cumulative returns surpass those of Gold, its price dynamics resemble those of a high-growth technology stock rather than a traditional safe-haven asset, calling into question its suitability as "Digital Gold." Gold's consistent performance, on the other hand, reinforces its function as a conservative vehicle for capital preservation, especially during periods of market uncertainty. Bitcoin also exhibits significantly higher volatility compared to Gold and the S&P 500, reinforcing its risk-asset characteristics.

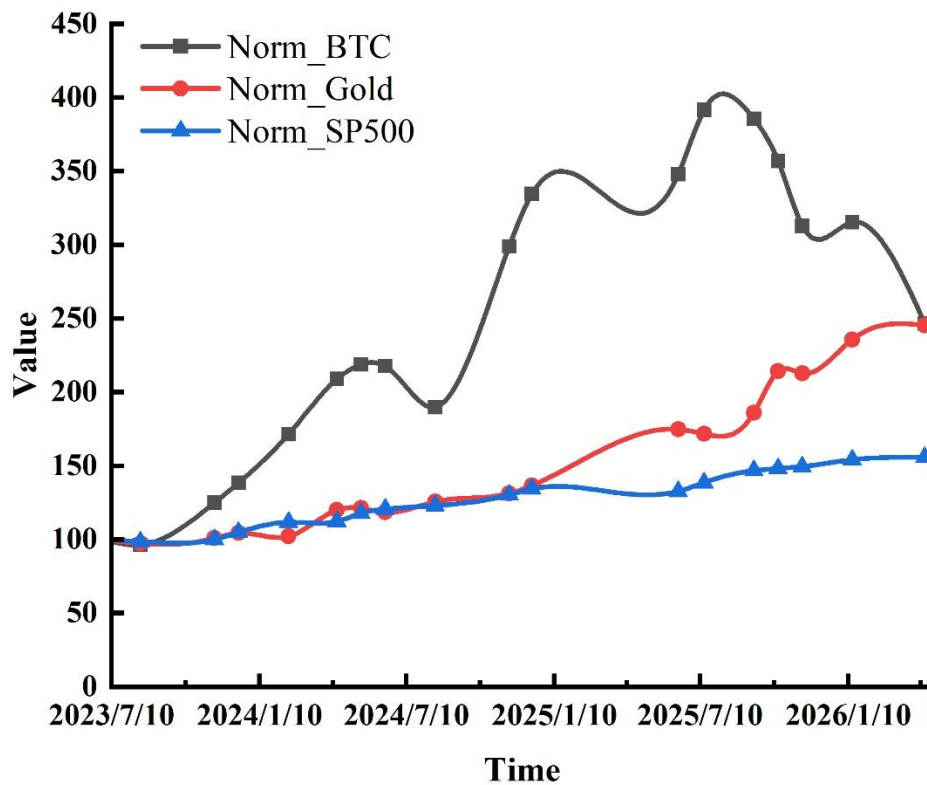


Figure 1. Comparative Performance of Bitcoin, Gold, and S&P 500.

4.2 Safe-Haven Analysis Based on DCC-GARCH

To capture the time-varying nature of asset correlations under different market conditions, this study employs a DCC-GARCH framework to estimate the dynamic conditional correlations between Bitcoin, Gold, and the S&P 500. Figure 2 presents the resulting time-varying correlations, with shaded regions indicating periods of elevated market stress ($VIX > 18$). The results reveal a clear divergence between Bitcoin and Gold in their behavior during turbulent market conditions. Bitcoin exhibits a relatively high and time-varying positive correlation with the equity market, typically ranging between 0.2 and 0.5. Notably, during high-stress periods, the correlation increases substantially, in some instances exceeding 0.6. This indicates that Bitcoin behaves as a pro-cyclical asset and fails to provide downside protection during market turmoil. In contrast, Gold maintains a

low and stable correlation with the S&P 500, generally fluctuating between 0.02 and 0.08. Even during periods of elevated volatility, its correlation remains close to zero or slightly negative, supporting its role as a traditional safe-haven asset. The divergence between the two assets becomes particularly pronounced during high-stress regimes, highlighting the fundamentally different roles of Gold and Bitcoin in portfolio risk management.

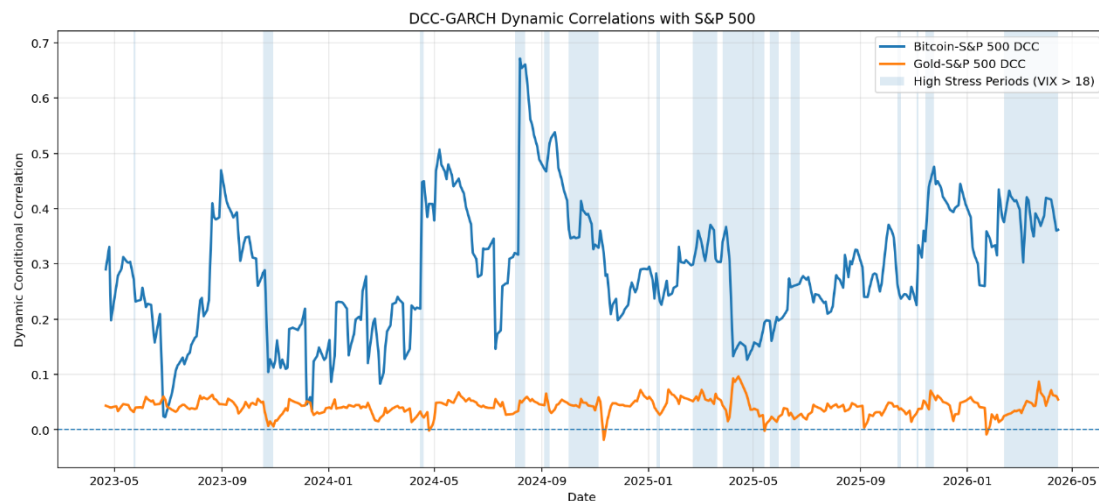


Figure 2. DCC-GARCH Dynamic Conditional Correlations between Bitcoin, Gold, and the S&P 500 (Shaded areas indicate VIX > 18).

4.3 Hedging against Inflation Expectations

Figure 3 presents the co-movement between Gold and the 5-Year, 5-Year Forward Inflation Expectation Rate. Figure 3 provides a visual illustration based on price levels, while the regression analysis is conducted using returns and changes in inflation expectations. The figure shows that Gold continued to rise during 2025–2026, while inflation expectations generally declined from earlier peaks. This pattern suggests that Gold's performance during this period was not driven by a simple positive relationship with inflation expectations. Instead, its price movement appears to reflect a broader combination of factors, including real interest rates, monetary policy expectations, and demand for defensive assets.

To further assess inflation-hedging effectiveness, this study estimates the relationship between asset returns and changes in inflation expectations using regression analysis. The results indicate that both Bitcoin and Gold exhibit very weak sensitivity to short-term fluctuations in inflation expectations. Specifically, Bitcoin shows a coefficient of 0.003, while Gold exhibits a slightly negative coefficient of -0.029. These near-zero coefficients suggest that neither asset responds strongly to daily changes in forward-looking inflation expectations. Bitcoin's minimal positive

sensitivity indicates that its price dynamics are not directly driven by inflation expectations but are more likely influenced by broader liquidity conditions and market sentiment. Meanwhile, the slightly negative coefficient for Gold reflects the dominance of the real interest rate effect, where rising inflation expectations may be associated with tighter monetary policy, thereby increasing the opportunity cost of holding non-yielding assets.

The results suggest that neither asset provides a simple and stable short-term hedge against inflation expectations. Bitcoin shows greater sensitivity but lacks stability, while Gold behaves more as a long-term store of value and defensive asset than as a direct short-term inflation hedge.

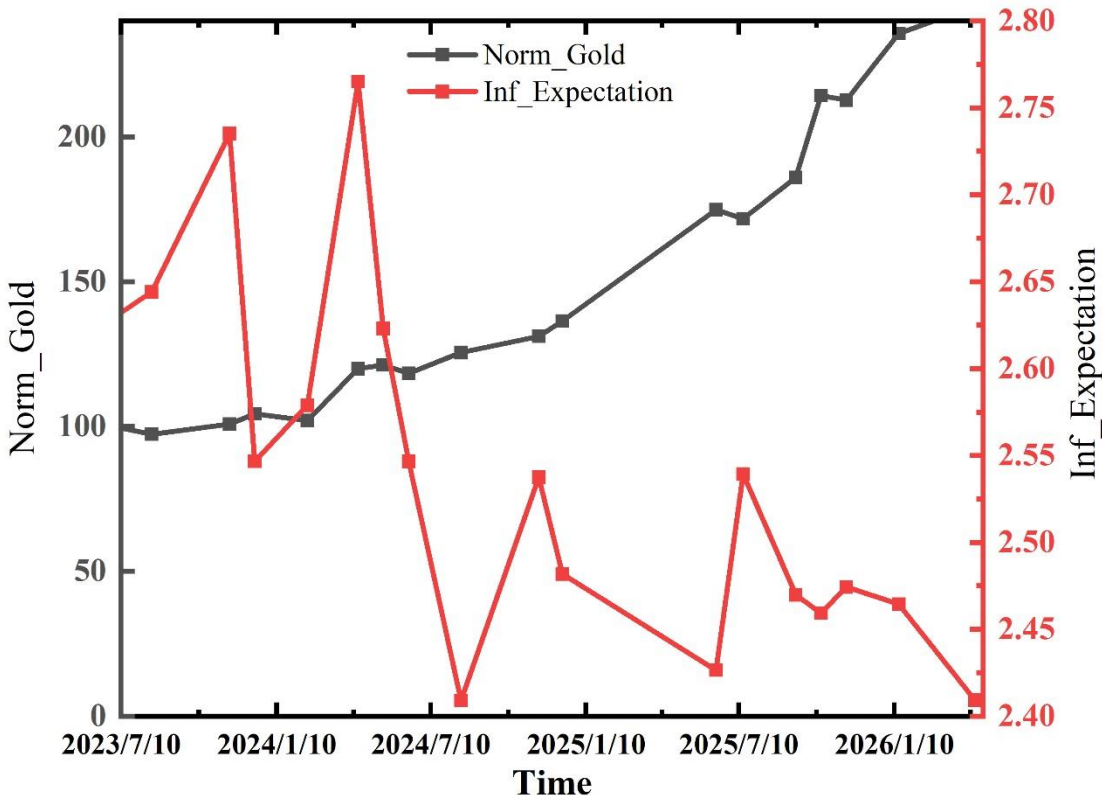


Figure 3. Gold Price and Inflation Expectations (T5YIFR).

5. Discussion

The empirical findings provide a nuanced understanding of the distinct roles that Gold and Bitcoin play in contemporary portfolios. Although both are often categorized as “alternative assets,” their behavior during the 2023–2026 period highlights fundamental differences in risk exposure, return characteristics, and hedging effectiveness.

5.1 The Deconstruction of “Digital Gold”

A key observation is Bitcoin's inability to function as a reliable safe-haven during systemic market shocks. As shown in Figure 2, Bitcoin's correlation with the S&P 500 rises above 0.3–0.6

during periods of elevated volatility, indicating its integration into the global liquidity cycle. This pattern aligns with the “Institutionalization Hypothesis”: following the approval of spot ETFs and the influx of large-scale institutional capital, Bitcoin is increasingly treated as a high-beta risk asset. During market panics, institutional investors may liquidate highly liquid assets—including Bitcoin—to meet margin calls or reduce portfolio Value at Risk. Consequently, the “Digital Gold” narrative appears more as a long-term thematic aspiration rather than a reliable short-term defensive strategy.

5.2 Gold's Paradox: Inflation vs. Opportunity Cost

The weak negative correlation between Gold and the 5Y5Y Inflation Swap observed in Figure 3 challenges the simplistic view of Gold as a linear hedge against inflation. During the 2025–2026 period, the “Real Interest Rate Effect” dominated Gold's price dynamics. In the current macroeconomic environment, rising inflation expectations were frequently countered by hawkish monetary signals, leading to higher-for-longer interest rates. As a non-yielding asset, Gold's opportunity cost increased under these conditions, suppressing its price despite elevated inflation concerns. Therefore, Gold's value during this period was driven more by its role as a geopolitical and systemic insurance instrument than by short-term consumer price inflation.

5.3 Portfolio Implications in a Polarized Market

The divergence between Bitcoin and Gold offers important insights for portfolio construction. Gold remains the “true” safe haven, effectively insulating portfolios during periods of equity market stress. Bitcoin, while lacking defensive properties during market downturns, provides asymmetric upside potential during recovery phases and periods of liquidity expansion. These results suggest that a binary choice between Gold and Bitcoin is suboptimal. Instead, a dual-allocation strategy enables investors to capture Bitcoin's growth and scarcity premium while relying on Gold to dampen portfolio volatility during high-stress periods, as highlighted in the shaded regions of Figure 2.

6. Conclusion

This study provides a comprehensive assessment of Gold and Bitcoin as safe-haven assets and inflation hedges during the volatile 2023–2026 period. The empirical evidence highlights a clear divergence in their roles. Gold consistently demonstrates the characteristics of a “true” safe haven, maintaining low and stable correlations with equities during periods of market stress and effectively

insulating portfolios against tail risks. Its relationship with long-term inflation expectations is nuanced: the negative correlation observed with the 5-Year, 5-Year Forward Inflation Expectation Rate reflects the predominance of the real interest rate effect, where rising central bank rates increase the opportunity cost of holding non-yielding assets. Bitcoin, in contrast, behaves as a pro-cyclical, high-beta asset: correlations with the S&P 500 intensify during high-volatility regimes, indicating that it does not provide defensive protection during systemic shocks. This pattern reflects the ongoing financialization of Bitcoin, driven by institutional participation and global liquidity cycles rather than its theoretical “Digital Gold” characteristics.

From a portfolio perspective, these findings imply that Gold remains indispensable for risk mitigation, while Bitcoin offers high-alpha potential during periods of liquidity expansion and market recovery. A dual-allocation strategy, leveraging Gold's stability and Bitcoin's asymmetric upside, may provide an optimal balance between protection and growth, allowing investors to capture the technological and scarcity premium of Bitcoin while relying on Gold to dampen volatility during stress periods.

Nonetheless, the rapidly evolving regulatory landscape for crypto-assets introduces significant uncertainties. Future research could explore the effects of sovereign-level Bitcoin adoption or the development of stablecoin-backed Gold instruments, which may bridge the gap between digital efficiency and physical security. Overall, while both assets are classified as alternative investments, their strategic functions are fundamentally distinct, necessitating differentiated approaches in risk management, portfolio allocation, and policy consideration.

Reference

- Acikgoz, T. (2025). Gold and Bitcoin as hedgers and safe havens: Perspective from nonlinear dynamics. *Resources Policy*, 102, 105489. <https://doi.org/10.1016/j.resourpol.2025.105489>
- Choi, S., & Shin, J. (2022). Bitcoin: An inflation hedge but not a safe haven. *Finance Research Letters*, 46, 102379. <https://doi.org/10.1016/j.frl.2021.102379>
- Dwita Mariana, C., Ekaputra, I. A., & Husodo, Z. A. (2021). Are Bitcoin and Ethereum safe-havens for stocks during the COVID-19 pandemic? *Finance Research Letters*, 38, 101798. <https://doi.org/10.1016/j.frl.2020.101798>
- Julaiti, J., Li, Y., & Song, K. (2026). Gold vs. “digital gold”: Bitcoin’s market positioning in the Trump 2.0 era. *Finance Research Letters*, 88, 109189.

<https://doi.org/https://doi.org/10.1016/j.frl.2025.109189>

Li, F., & Yin, Y. (2026). Study on the safe-haven and hedging roles of bitcoin, gold, and crude oil on global stock markets in short-term, medium-to-long-term, and shock periods. *Economic Analysis and Policy*, 90, 472-489. <https://doi.org/https://doi.org/10.1016/j.eap.2026.01.043>

Snene Manzli, Y., & Jeribi, A. (2024). Assessing Bitcoin, gold and gold-backed cryptocurrencies as safe havens for energy and agricultural commodities: insights from COVID-19, Russia–Ukraine conflict and SVB collapse. *Journal of Financial Economic Policy*, 16(5), 656-689. <https://doi.org/https://doi.org/10.1108/JFEP-12-2023-0386>

Teruel-Gutiérrez, R., Conesa, J. A. B., & Serrano, G. A. (2026). Stress selective Behavior: Asymmetric Responses of Bitcoin and Gold to Valuation and Uncertainty Shocks. *Finance Research Letters*, 110036. <https://doi.org/https://doi.org/10.1016/j.frl.2026.110036>

Wen, F., Tong, X., & Ren, X. (2022). Gold or Bitcoin, which is the safe haven during the COVID-19 pandemic? *International Review of Financial Analysis*, 81, 102121. <https://doi.org/https://doi.org/10.1016/j.irfa.2022.102121>